

KAZI EHSANUL KARIM

Molecular Simulation | Nanoscale Transport | Multiscale Thermo-Fluids | Machine-Learning Interatomic Potentials

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Postdoctoral / Research Fellow Opportunities | Target start: early 2027

RESEARCH PROFILE

Computational researcher and Ph.D. candidate in Mechanical Engineering at the University of Ulsan, with first-author publications in Physical Review E and Physical Chemistry Chemical Physics. Combines molecular dynamics, statistical mechanics and continuum modelling to investigate mass and energy transport under nanoscale confinement. Current research connects CP2K reference calculations, DeePMD models and LAMMPS simulations for CNT-confined NaCl–water, focusing on model-error diagnosis and targeted reference data acquisition. Research interests include error-guided active learning, physics-informed modelling and agentic workflows that allocate computation using physical and statistical evidence.

Research methods: LAMMPS EMD/NEMD; CP2K; DP-GEN / DeePMD-kit; Green–Kubo transport; continuum boundary conditions; Python / Linux / HPC; ANSYS thermo-fluids.

EDUCATION

University of Ulsan, Republic of Korea

Mar 2019 – Present

Ph.D. Candidate, Integrated M.S.–Ph.D. in Mechanical Engineering | Micro/Nano-scale Thermo-Fluids Engineering

Laboratory | Supervisor: Prof. BoHung Kim

Dissertation defense planned: November 2026 | Formal degree conferral expected: February 2027

Dissertation: What Survives Sub-nm Confinement? — The Meaning of Slip Length under Molecular Accessibility, Ergodicity, and Structure–Transport Decoupling.

Khulna University of Engineering & Technology (KUET), Bangladesh

May 2015

B.Sc., Mechanical Engineering

DOCTORAL RESEARCH EXPERIENCE

Doctoral Researcher — Molecular Simulation & Nanofluidics, University of Ulsan

Mar 2019 – Present

- **Boundary thermodynamics** — Developed a first-law energy-balance analysis of the nanoscale flow boundary and tested NEMD against Hagen–Poiseuille limits under controlled wall–fluid interaction (Physical Review E, 2021).
- **Water transport in CNTs** — Critically assessed scale effects and boundary definitions in CNT water-transport models, relating molecular-simulation evidence to continuum predictions (Physical Chemistry Chemical Physics, 2024).
- **Confinement and transport** — Investigate molecular accessibility, layering, finite-time diffusivity, hydraulic resistance and continuum-closure limits in carbon nanotubes and graphene nanoslits.
- **Reproducibility and uncertainty** — Apply ensemble, timestep, box-size, forcing and sampling checks; block averaging and confidence intervals; and thermostat-sensitivity and cross-method comparisons to test transport interpretations.

TECHNICAL EXPERTISE

- **Molecular simulation** — LAMMPS; EMD/NEMD; NVT/NPT/NVE; confined fluids; pressure-driven and Couette flow; Green–Kubo and NEMD transport coefficients; interfacial friction/slip; spatial binning and profile analysis.
- **CFD & continuum modelling** — ANSYS geometry/mesh generation, compressible nozzle flow, boundary-condition specification, convergence monitoring, and pressure/density/temperature/velocity/Mach post-processing; nozzle-wall and coolant-passage thermal analysis.
- **Scientific computing & statistics** — Python (NumPy, SciPy, pandas, Matplotlib, MDAnalysis, scikit-learn), Bash, Linux/HPC, MPI workflows, automated parameter sweeps; block averaging, confidence intervals, convergence/sensitivity diagnostics, uncertainty-aware regression.
- **Electronic structure & MLIP** — CP2K periodic GPW reference calculations using PBE-D3(BJ), GTH pseudopotentials and DZVP-MOLOPT basis sets; DP-GEN active learning; DeePMD-kit dataset curation, committee training, holdout validation, error analysis, stability testing and LAMMPS deployment.

INDEPENDENT & MULTISCALE PROJECTS

Error-Aware MLIP Workflow for CNT-Confined Electrolytes

2026 – Present

- **Workflow** — Developing an end-to-end CP2K → DP-GEN → DeePMD → LAMMPS workflow for confined ionic systems.
- **Validation** — Using independent holdout calculations, element-resolved force-error analysis, committee disagreement and stability testing to identify under-resolved configurations and guide targeted data acquisition.
- **Objective** — Enable auditable MLIP deployment for transport simulations while detecting conditions that require additional first-principles information.

Thermo-Fluid Simulation of a Methane Rocket-Engine Nozzle & Cooling Passage

2022

Industry-linked laboratory project using ANSYS for geometry/mesh generation, compressible nozzle-flow and high-temperature thermal analysis, and post-processing of density, pressure, temperature, velocity, and Mach-number fields; included nozzle-wall and methane coolant-passage thermal behaviour.

Research Assistant — Robotics & Mechatronics Laboratory, KUET

Feb 2014 – May 2015

Kinematic modelling of an RPR manipulator, automated-wheelchair design, and experimental thermal/materials studies.

PUBLICATIONS & MANUSCRIPTS

Peer-reviewed journal articles

1. K. E. Karim, M. Barisik, C. Bakli, B. Kim, “Estimating water transport in carbon nanotubes: A critical review and inclusion of scale effects,” *Physical Chemistry Chemical Physics*, 2024, 26(28), 19069–19082. [DOI](#)
2. K. E. Karim, B. Kim, “First law of thermodynamics on the boundary for flow through a carbon nanotube,” *Physical Review E*, 2021, 103(5), 053115. [DOI](#)

Under review / completed

3. K. E. Karim, B. Kim, “Molecular admission and packing govern transport in subnanometre graphene slits,” under review.
4. K. E. Karim, B. Kim, “Wall–fluid interaction strength shifts the slit width of maximum finite-time diffusivity in a graphene-confined simple fluid,” under review.
5. K. E. Karim, B. Kim, “Confinement-Dependent Hydraulic Resistance and Limits of Navier-Slip Transferability in Graphene Nanoslits,” manuscript in final preparation.

In preparation — current research directions

Physics-informed neural networks (PINNs) for structure–friction–flow closure; multimodal electrolyte transport in carbon nanotubes; and error-aware machine-learning interatomic potentials (MLIPs) for agentic multiscale simulation.

Additional peer-reviewed work

- A. S. M. M. Rahman, K. E. Karim, M. H. S. Simanto, “Effect of heat treatment on low carbon steel: An experimental investigation,” *Applied Mechanics and Materials*, 2016, 860, 7–12. [DOI](#)

Also: one additional journal article on RPR manipulator modelling and five peer-reviewed conference papers in thermal, materials, and design engineering.

REFERENCES

References available upon request.